Initials:_____

ANSWER KEY

BME 105: Genetics in the Genomics Era
Winter 2020: Midterm 1

Name: _____ANSWER KEY_____

Student ID:_____

Before starting, write your initials on the top of each page.

You can use the back-side of the pages for scratch, but we will not grade answers written on the back-side of the page.

Section	Points
Part I: One, two, or three-word answers and T/F	/16
Part II: Multiple choice	/18
Part III: Short answer	/16
Part IV: Extended calculation	/50
Total Points	**/100**

Second Letter

1st letter	U		C		A		G		3rd letter
U	UUU UUC UUA UUG	Phe Leu	UCU UCC UCA UCG	Ser	UAU UAC UAA UAG	Tyr Stop Stop	UGU UGC UGA UGG	Cys Stop Trp	U C A G
C	CUU CUC CUA CUG	Leu	CCU CCC CCA CCG	Pro	CAU CAC CAA CAG	His Gln	CGU CGC CGA CGG	Arg	U C A G
A	AUU AUC AUA AUG	Ile Met	ACU ACC ACA ACG	Thr	AAU AAC AAA AAG	Asn Lys	AGU AGC AGA AGG	Ser Arg	U C A G
G	GUU GUC GUA GUG	Val	GCU GCC GCA GCG	Ala	GAU GAC GAA GAG	Asp Glu	GGU GGC GGA GGG	Gly	U C A G

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Part I: One, two, or three-word answers and True/False

In the space provided next to each definition, clearly write the appropriate term. Each answer is worth 2 points, unless otherwise specified.

1. _____sum rule_____ if two or more events are mutually exclusive, you can use this rule to determine the probability of the events occurring together

2. _____diploid_____ A cell with 2N chromosomes

3. single nucleotide polymorphism the abbreviation is SNP

4. _____somatic cells_____ cells that undergo mitosis

5. _____indel_____ a type of genetic variant that involves and insertion or deletion of about 1-100bp of DNA

6. _____anaphase I_____ the phase of meiosis where homologous chromosomes are pulled apart

7. Indicate if the following statements are True or False:
   a. T F For SSR variants, a microarray method for genotyping would be a preferred method.

      F

   b. T F Unlinked genes will have a recombination frequency of 50%

      T

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Part II: Multiple choice. Circle your answer. <u>If there is more than one correct answer, circle all correct answers.</u> Each question is worth 2 points, unless otherwise specified.

1. Sickle cell anemia is a recessive trait in humans. In a cross between a father who has sickle cell anemia and a mother who is heterozygous for the sickle cell allele, what is the probability that all of their first two children will be unaffected?
   a. 1/4

   b. 1/2

   c. None

   d. 1/8

   e. 1/16

   a

2. How many types of gametes will an individual of the genotype: AaBBCCDdEeFFGg produce?

   a. 2
   b. 4
   c. 8
   d. 16
   e. 32

   d

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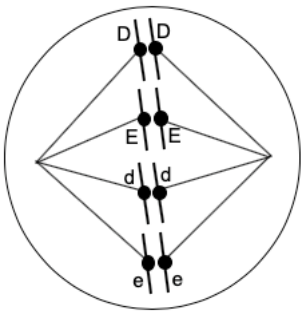
3. The following is a D/d;E/e cell in mitosis.



Is this organism:
   a. 2n=2
   b. 2n=4
   c. 4n=4
   d. 4n=8

b

4. In a population of gametes from an AF/af individual, 15% of chromosome 7s are Af. How far apart (in m.u.) are the A and F genes?

   a. 7.5 m.u.

   b. 15 m.u.

   c. 30 m.u.

   d. 35 m.u.

   e. 60 m.u.

c

5. In X chromosome linked recessive traits, the phenotype is _____.
   a. always expressed in females
   b. always expressed in males
   c. never expressed in females
   d. never expressed in males

   b

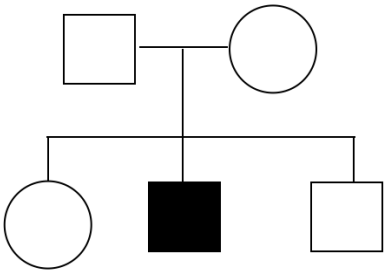
6. You receive a snippet of DNA sequence that contains the start of a protein sequence. How many possible protein sequences could this sequence contain?

   5' AATTAAGCATGGCAGATTATGACCCTAGATGCCGACCT 3'

   a. 1
   b. 2
   c. 3
   d. 4
   e. 6

   d

7. In the following pedigree, what is the likely mode of inheritance?



a. Autosomal dominant

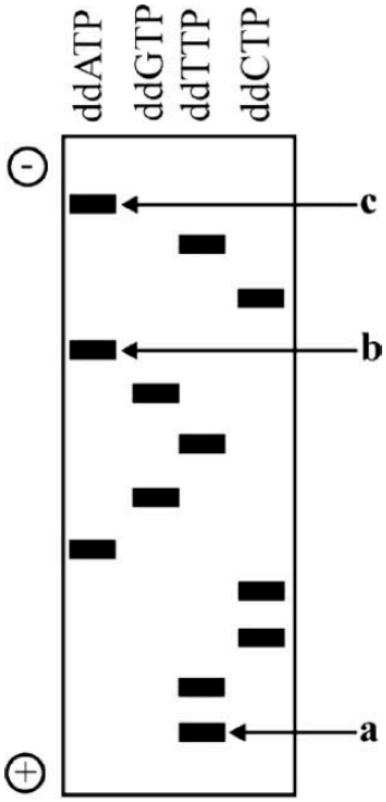
b. Autosomal recessive

c. Sex-linked dominant

d. Sex-linked recessive

b, d

Consider the following figure which shows a fictitious experiment of DNA sequencing using the Sanger sequencing method and the resulting sequencing gel.



8. DNA molecules corresponding to band **a** contain cytosine.

a. The above statement is true

b. The above statement is false

c. There is not enough information.

c, the actual DNA molecule corresponding to the band would include the primer sequence, which is not given, plus the extended nucleotides from the dNTPs and ddNTPs.

9. DNA molecules corresponding to band **b** contain cytosine

    a. The above statement is true

    b. The above statement is false

    c. There is not enough information

    a

Part III: Short answer. Each answer is worth 2 points, unless otherwise specified. No partial credit given, unless specified.

1. Regarding the sequencing gel above, what is the sequence of DNA as determined by this Sanger sequencing method? Indicate 5' and 3' end of the sequence.

   5'-TTCCAGTGACTA-3'

2. You have two true-breeding varieties of elongated pumpkins (strain 1, and strain 2). True breeding wild-type pumpkin strains are round. When crossed to wildtype, strain 1 and strain 2 produce round pumpkins.

    a. Is the phenotype of strain 1 and strain 2 dominant or recessive to the phenotype of the wild-type strain?

       Recessive

    b. You cross true breeding strain 1 with true breeding strain 2 and find that all F1 pumpkins are elongated. After self-fertilization you find that all F2 pumpkins are elongated. What are the most likely genotypes of the parents, F1 and F2 plants? **The following answers are worth a total of 2 points (1/4 points for each answer).**

       Strain 1: _____ rr

       Strain 2: _____ rr

       F1: _____ rr

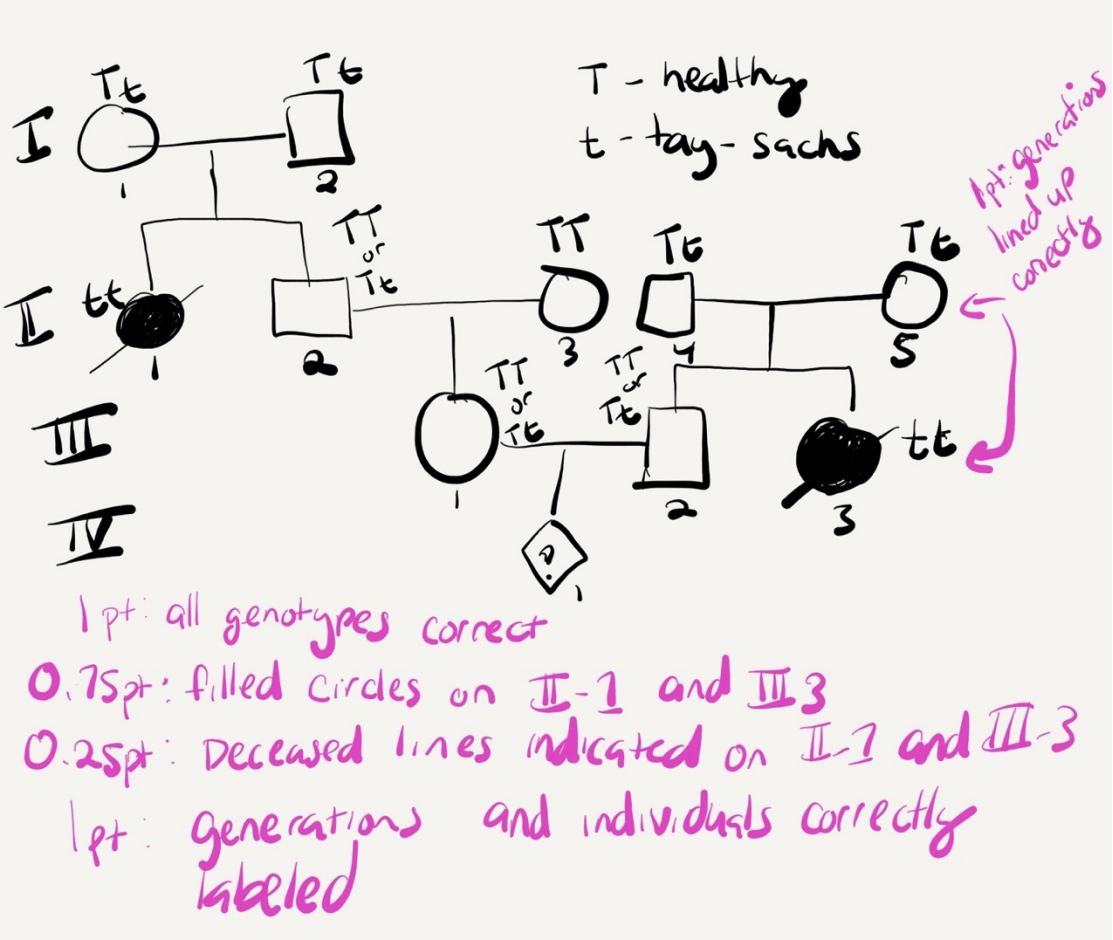
       F2: _____ rr

3. Triploid organisms (3N) are typically sterile. Why is that? Hint: Think about what mitosis and meiosis would look like for a triploid organism

   Somatic cells of triploid organisms can undergo mitosis, but they lack enough homologous chromosomes to pair up in meiosis.

4. Tay-Sachs is a recessive lethal disease. It leads to neurological deterioration soon after birth and death usually by the age of 4. This disease is rare in the population overall but is found at relatively high frequency in Ashkenazi Jews from Central Europe. A woman whose paternal aunt had the disease is trying to determine the probability that she and her husband could have an affected child. Her mother does not come from a high-risk population. Her husband's sister died of the disease at an early age.

Draw the pedigree of the individuals described. Include the genotypes, if known. If there are multiple possible genotypes for an individual, list all possible genotypes. Also number the generations and individuals. You can use pencil or blank sides of the exam as scratch work. **Your final pedigree has to be marked in pen and circled. This is worth 4 points. Partial credit will be given.**



5. A pregnant woman was recommended by her doctor to undergo carrier screening to get her genotype for a selection of autosomal-recessive and X-linked genetic conditions. The pregnant woman has had her whole-genome sequenced using high-throughput sequencing. For the following genetic conditions, can we determine her genotype from whole-genome sequencing. If not, why?
    a. Cystic fibrosis: Caused by SNP mutations in the *CFTR* gene.

       Yes. SNPs can be detected with high-throughput sequencing.

    b. Fragile X: Caused by mutations in the *FMR1* gene. Nearly all cases of fragile X syndrome are caused by a mutation in which a DNA segment, known as a CGG triplet repeat, is expanded to more than 200 times.

       Unlikely. High-throughput sequencing has a read length limitation (~150bp and some other modes have 300bp), making it difficult to determine repeat sizes that have longer lengths.

Part IV: Extended calculations (show your work for partial credit, show pedigrees and genotypes, punnett squares, etc.).

1. You have been hired as Brother Gregor's gardening assistant. He has several different patches of peas. Each patch is true-breeding and shows a single recessive trait. He points to his recessive white-flower patch (purple is dominant) and his recessive green-pea patch (yellow is dominant) and asks you to use plants from those patches to generate a true-breeding line that displays both white flowers and green peas. Explain how you would do it, or explain why it is impossible to do. **This is worth 25 points**.

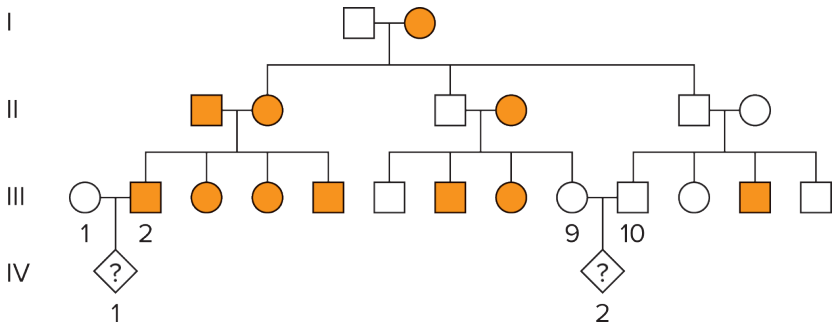
Y = yellow P = purple
y = green p = white

1) Mate white-flower (Y/Y; p/p) plants with green-pea (y/y; P/P) plants

2) The F1 offspring will be Y/y; P/p – self fertilize those

3) 1/16 of the F2 offspring will be white-flower and green-pea (y/y; p/p)

2. The following pedigree shows the inheritance of red hair in a family in Scotland. Red hair is caused by homozygosity for a recessive allele of a gene called MC1R. Although worldwide red hair is the rarest of human hair colors, red hair is not uncommon in Scotland. In fact, 40% of Scots without red hair are nonetheless carriers of the red hair allele.



a. Why does red hair show a horizontal inheritance pattern in this particular pedigree even though the trait is caused by a recessive allele? **This is worth 5 points.**

Recessive traits that are common in the population show a horizontal pattern of inheritance because recessive allele carriers and even homozygotes from outside the family will likely contribute gametes to every generation.

b. Assuming that individual III-2 has a child with the Scottish woman shown (III-1) who is not a close relative, what is the probability that this child (IV-1) will have red hair? **This is worth 10 points.**

Rr – brown hair or not red hair
rr – red hair
P(IV-1 is rr) = P(IV-1 is rr and III-1 is Rr)
= P(IV-1 is rr given III-1 is Rr) * P(III-1 is Rr)
= 0.5 *0.4 = 0.2 = 20%

c. What is the probability that the child of the first cousins III-9 and III-10 (IV-2) will have red hair? **This is worth 10 points.**

P(IV-2 is rr and III-9 is Rr and III-10 is Rr)
= P(IV-2 is rr given <u>III-9 is Rr and III-10 is Rr</u>) * P(<u>III-9 is Rr and III-10 is Rr</u>)
= P(IV-2 is rr given <u>III-9 is Rr and III-10 is Rr</u>)* P(III-9 is Rr) * P(III-10 is Rr)
= 1/4 * 1* 2/3 = 2/12 = 1/6